

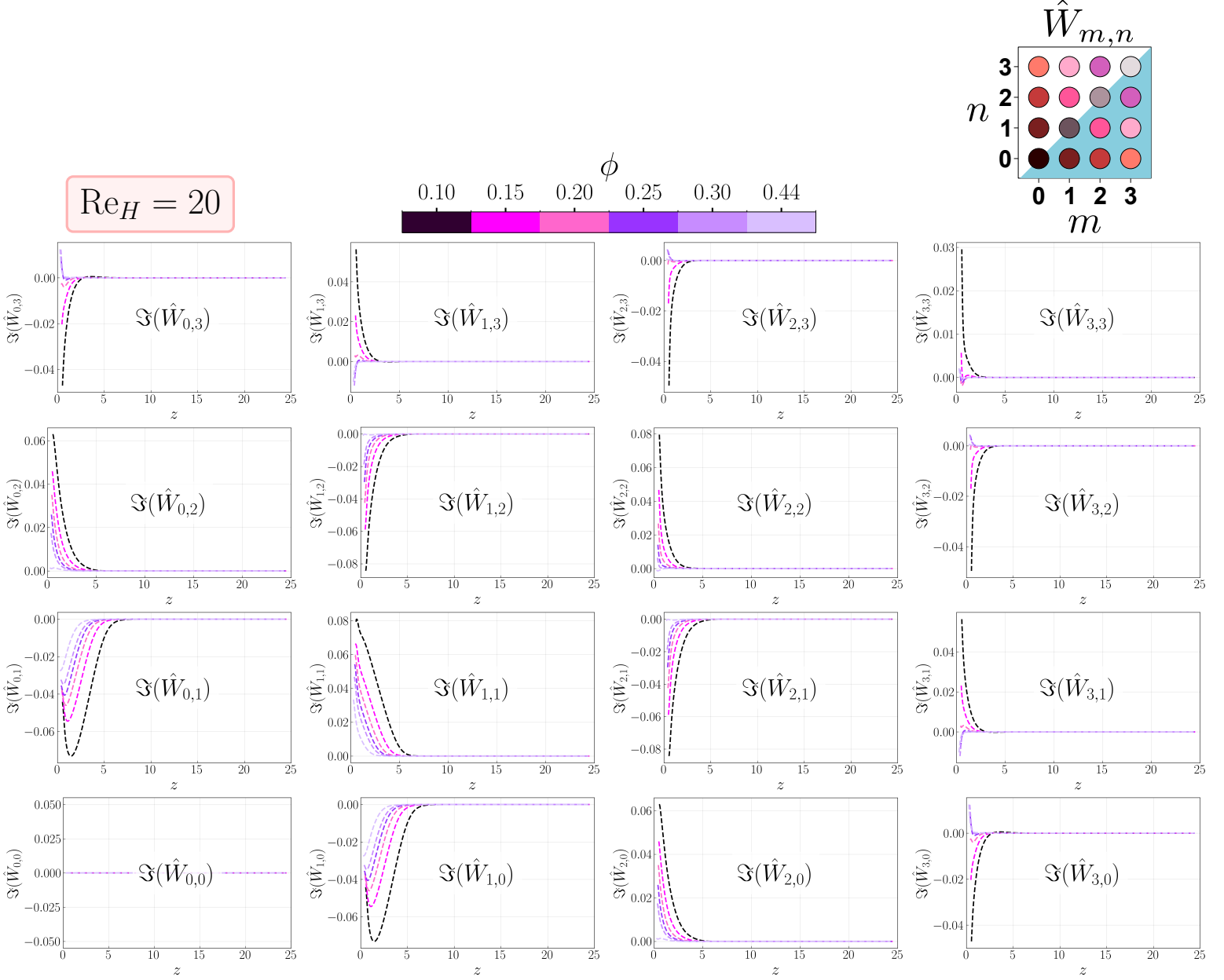


Figure B.14: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

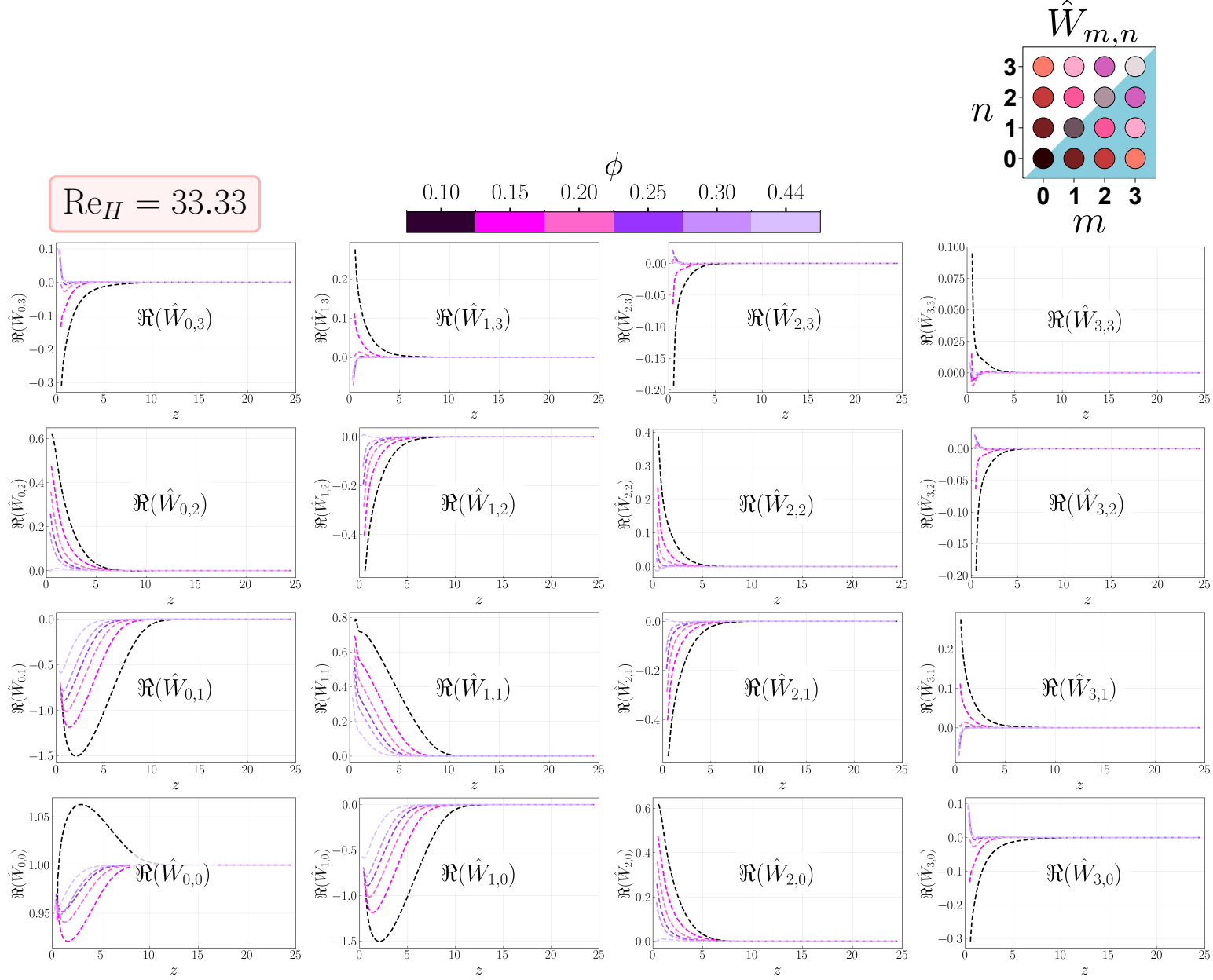


Figure B.15: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients down-stream of the SRG at $\text{Re}_H = 33.33$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

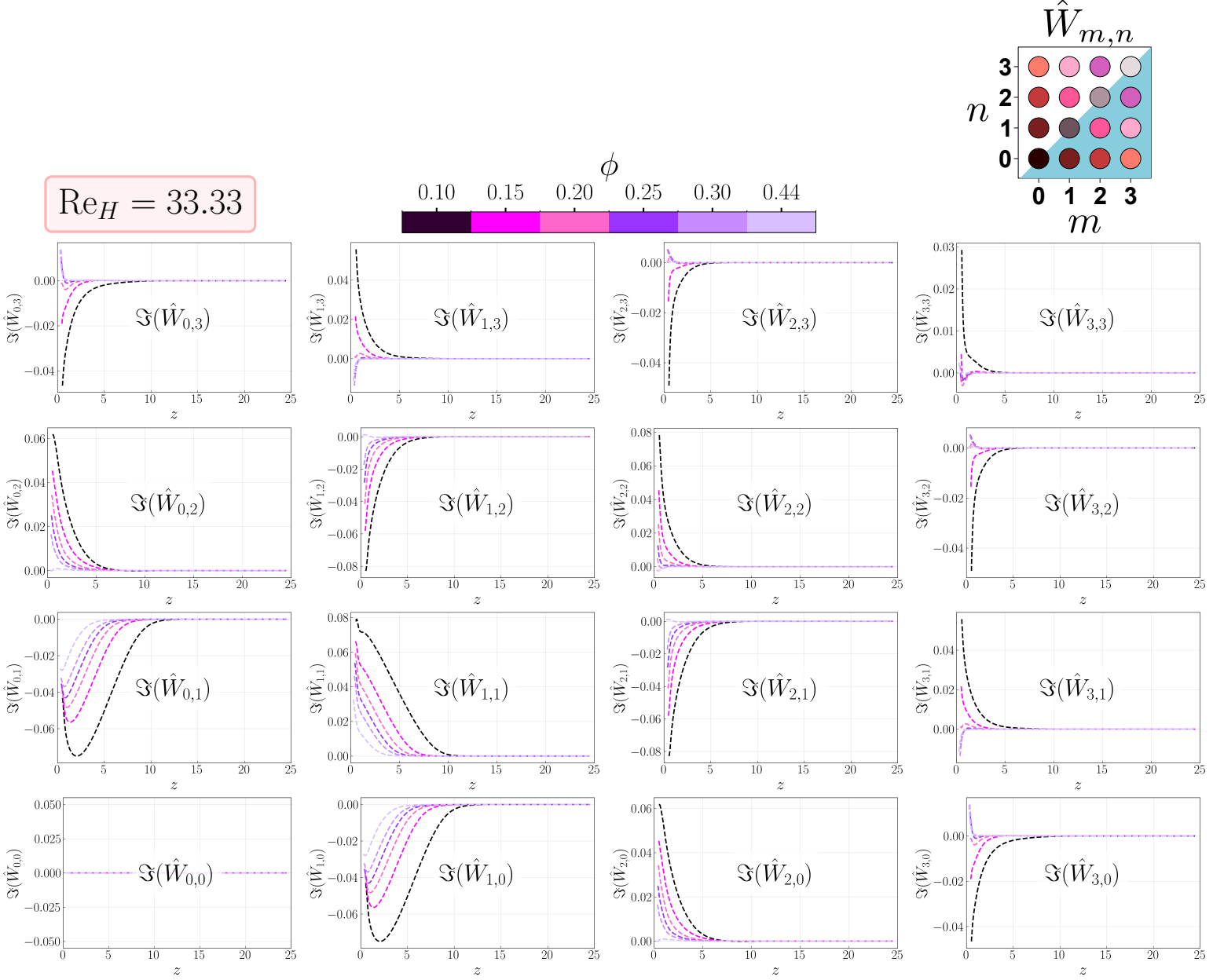


Figure B.16: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 33.33$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

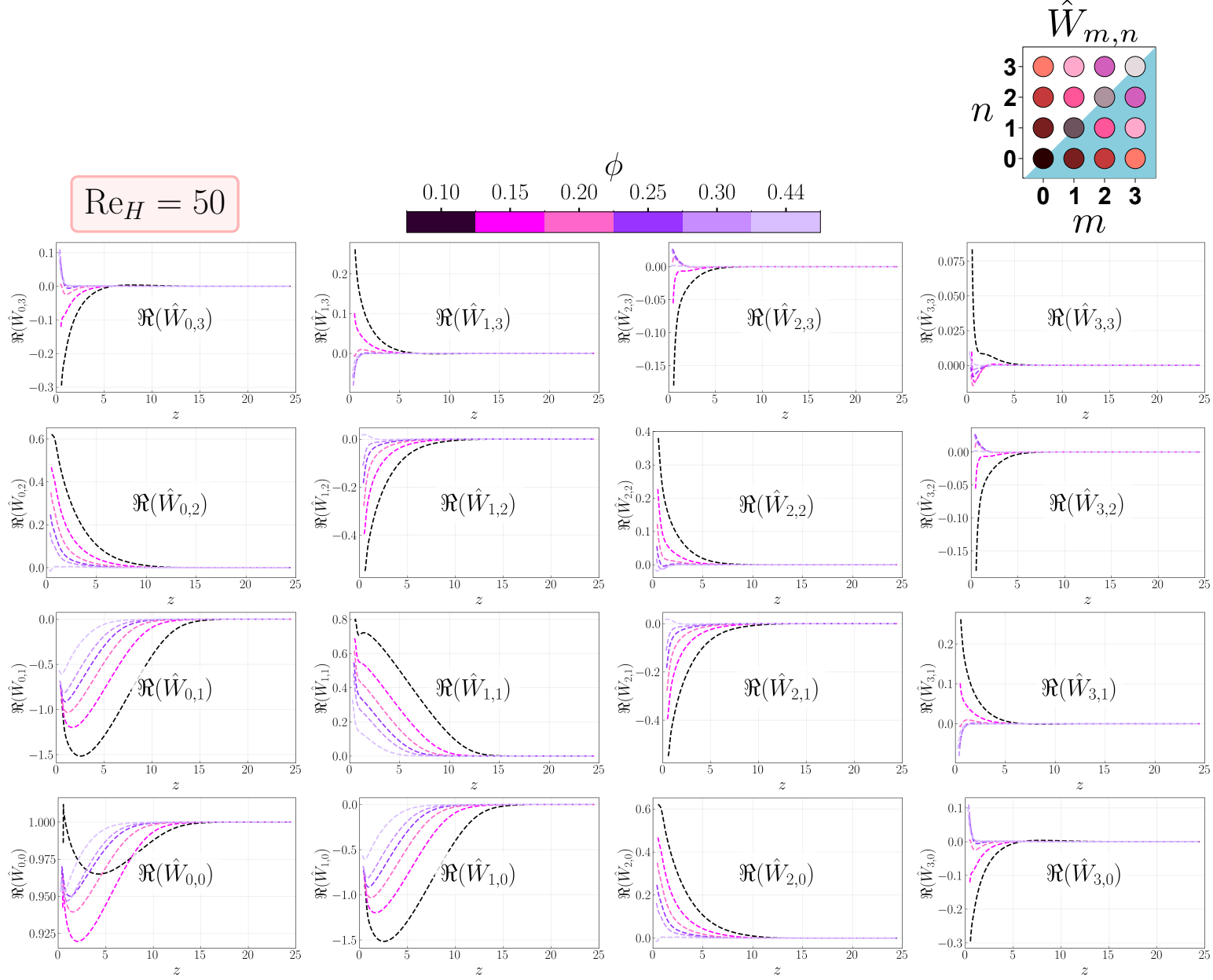


Figure B.17: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

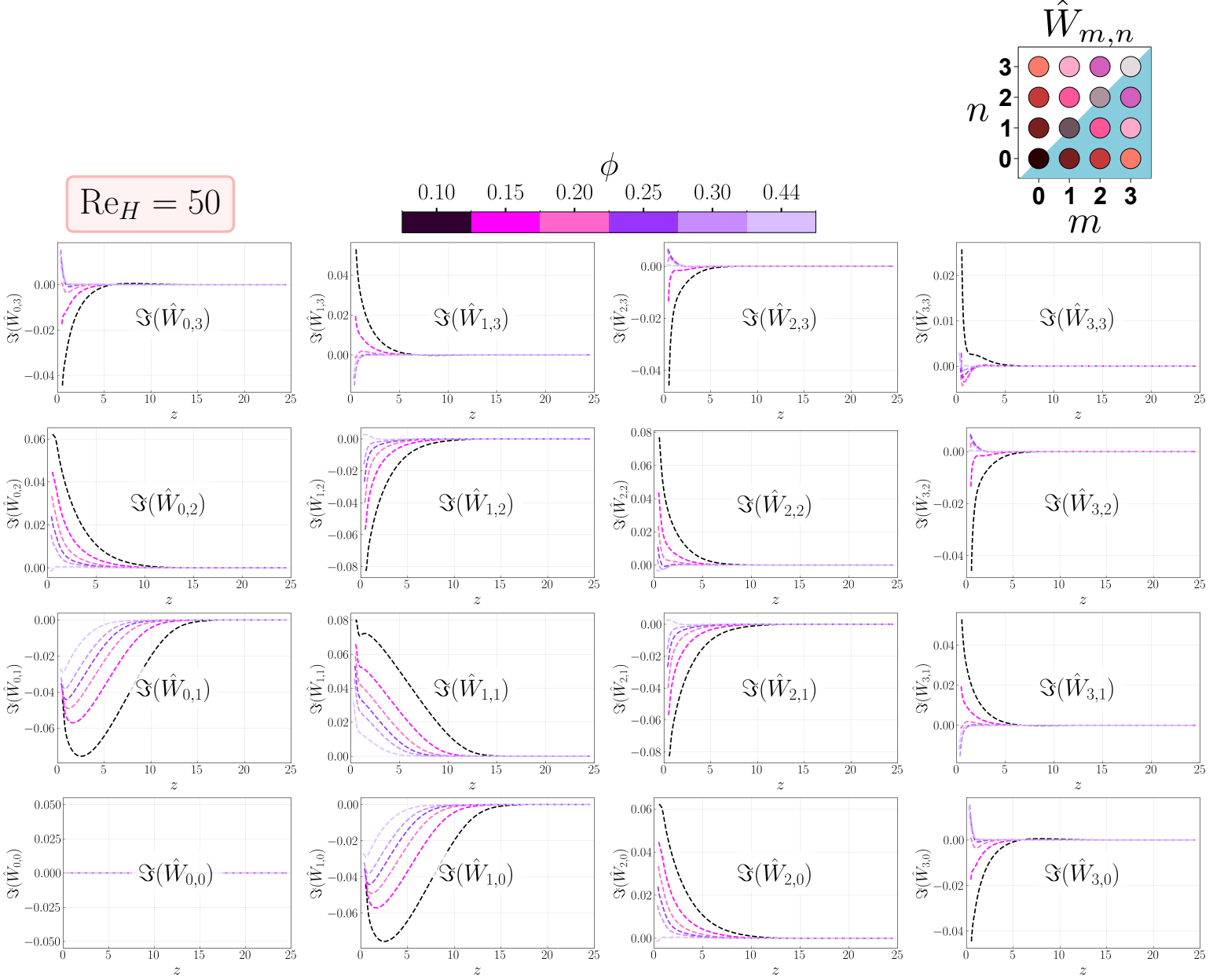


Figure B.18: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

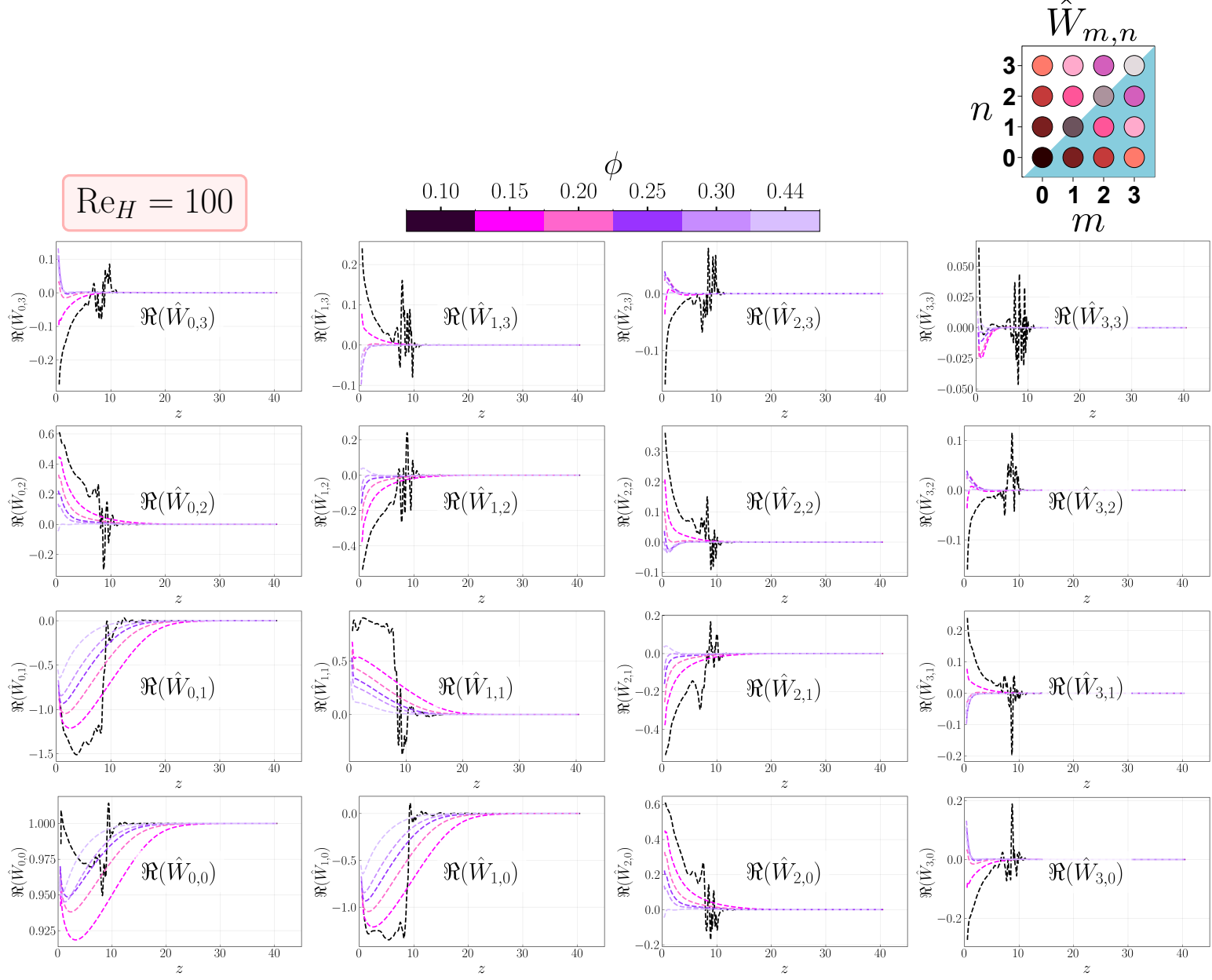


Figure B.19: Streamwise evolution of the real part $\Re(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 100$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the retained Fourier modes. For the steady-flow cases, the Fourier coefficients satisfy the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$; however, this symmetry is broken for the unsteady case at $\text{Re}_H = 100$ and $\phi = 0.10$.

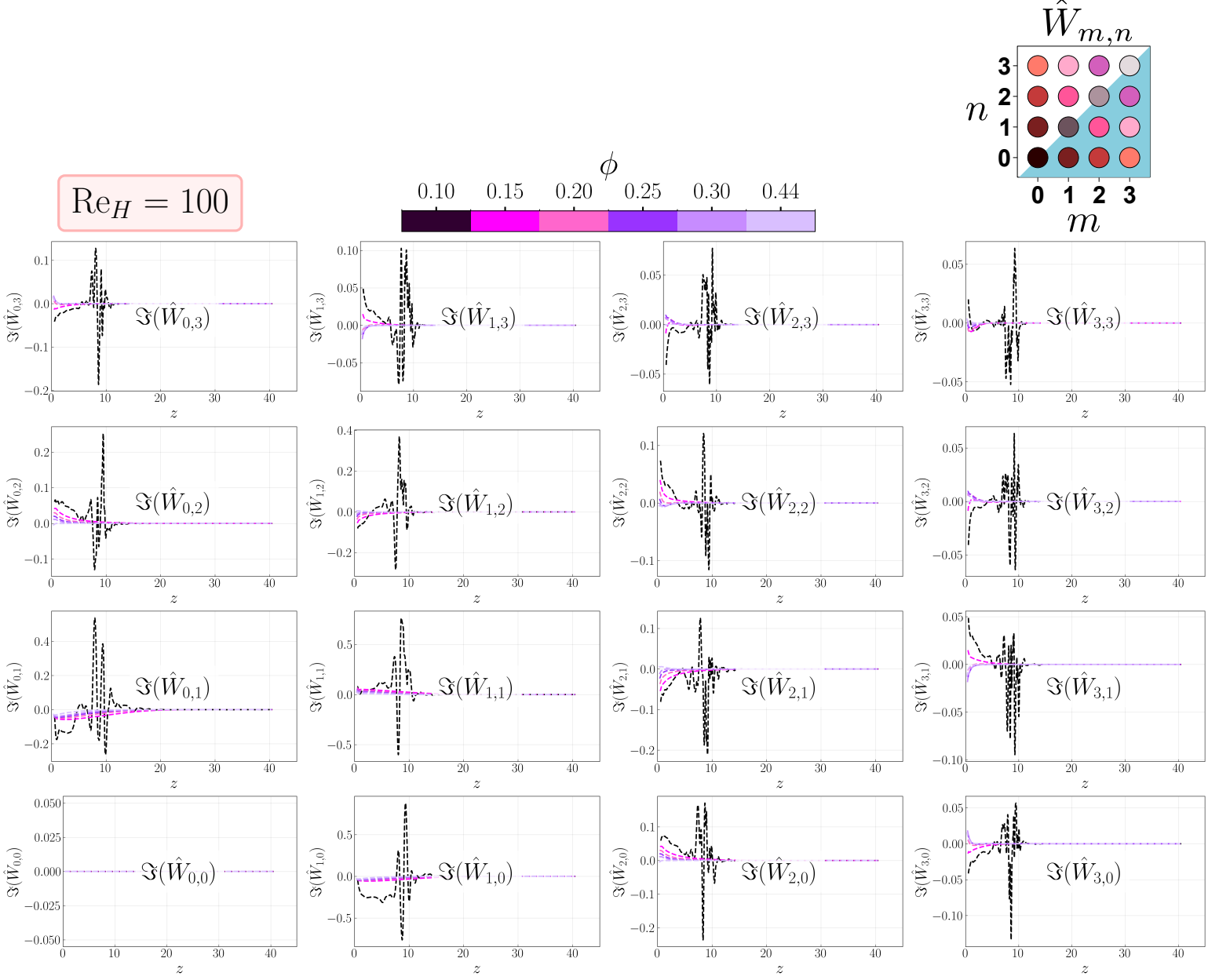


Figure B.20: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 100$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the retained Fourier modes. For the steady-flow cases, the Fourier coefficients satisfy the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$; however, this symmetry is broken for the unsteady case at $\text{Re}_H = 100$ and $\phi = 0.10$.

The following figures show the coefficients $\hat{V}_{m,n}$ of the transverse velocity component v .

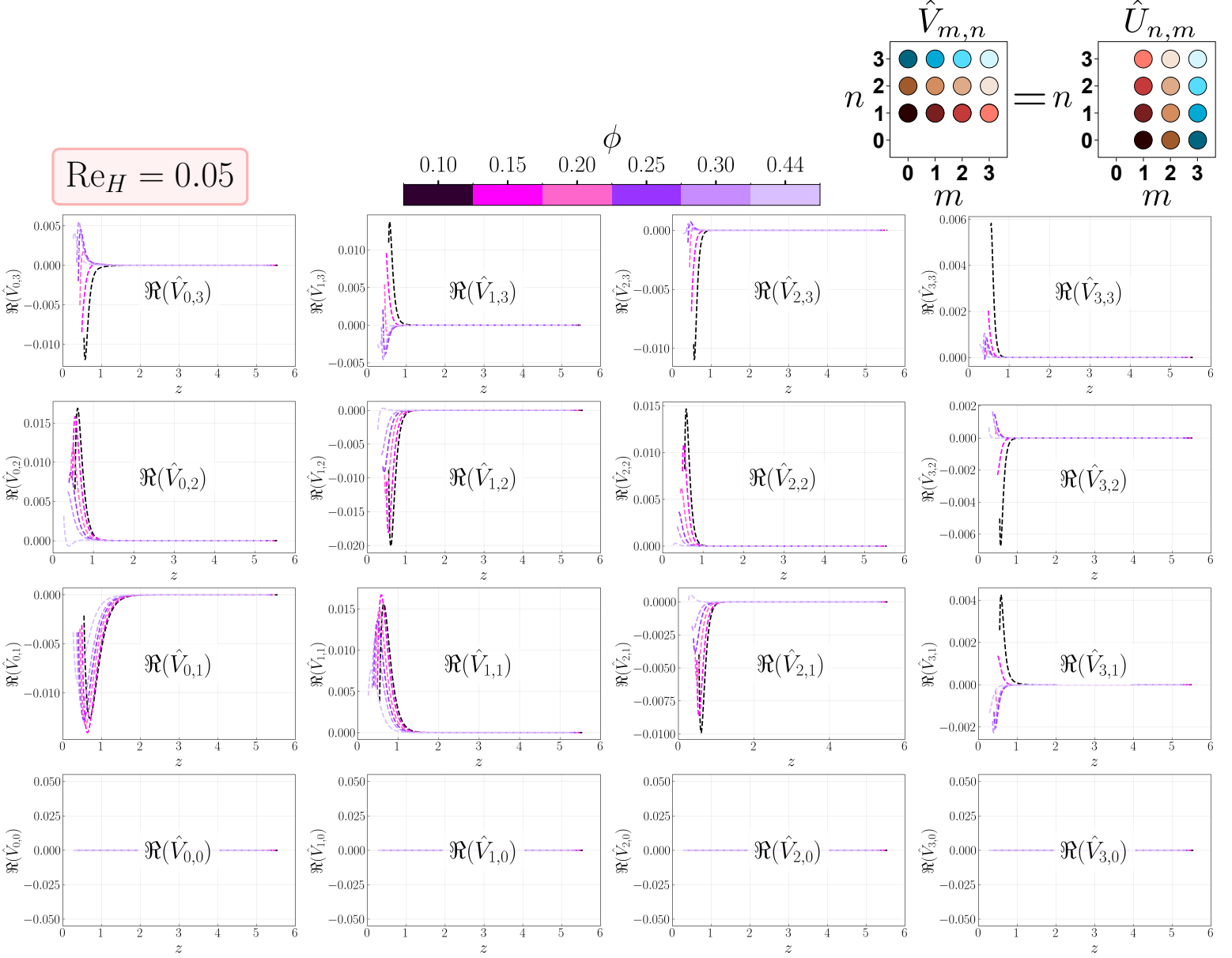


Figure B.21: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.05$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

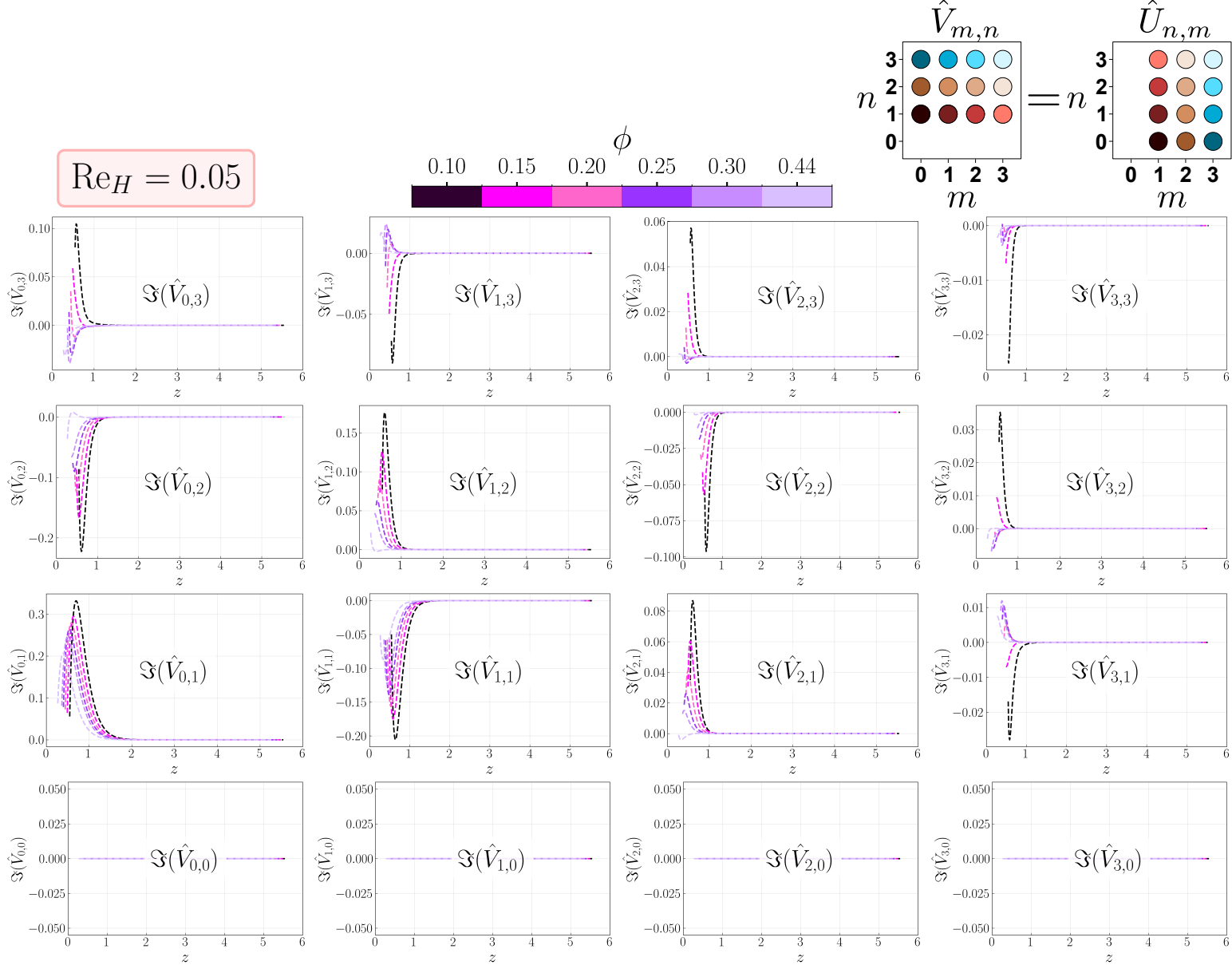


Figure B.22: Streamwise evolution of the imaginary part $\Im(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.05$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

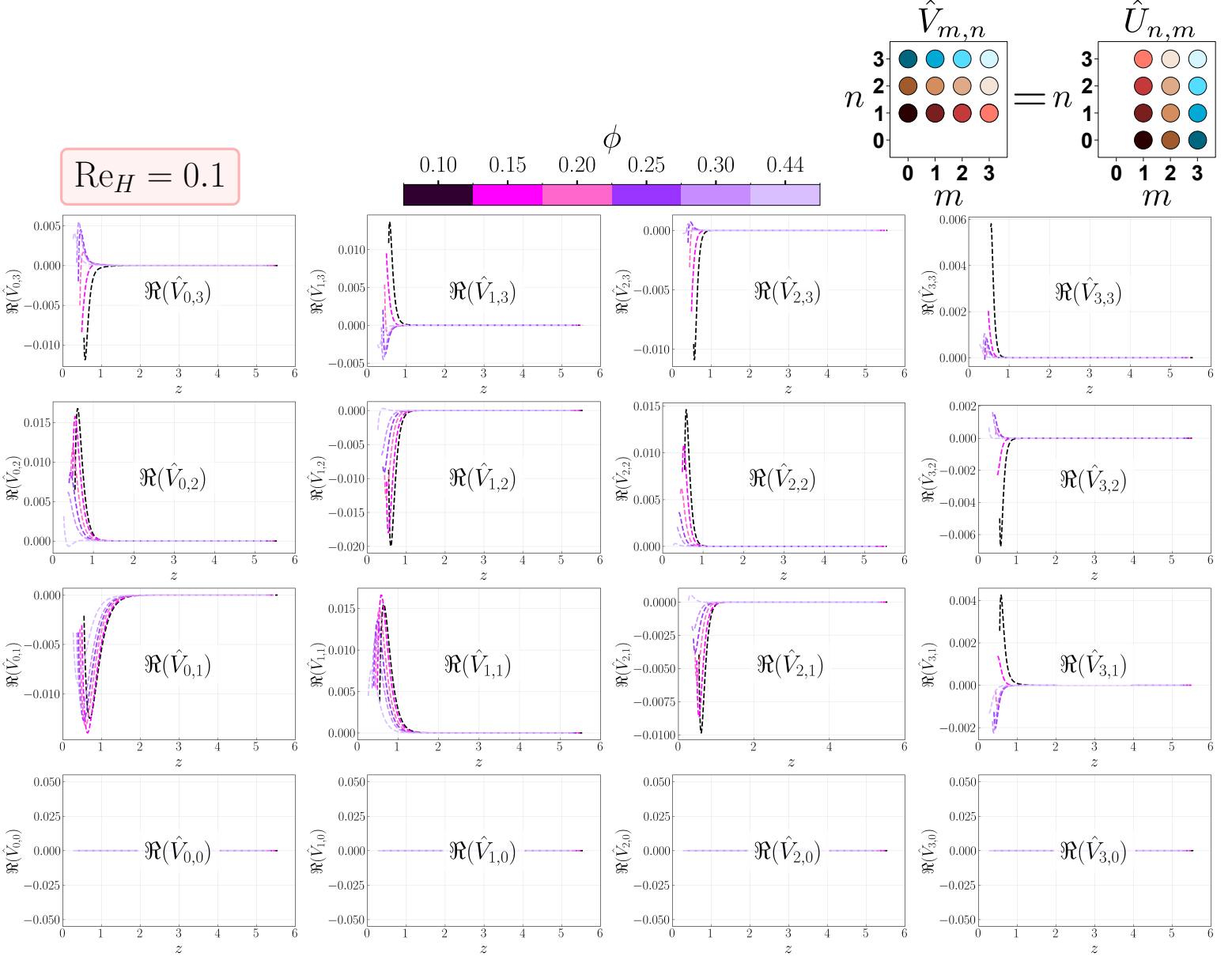


Figure B.23: Streamwise evolution of the real part $\mathcal{R}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.1$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

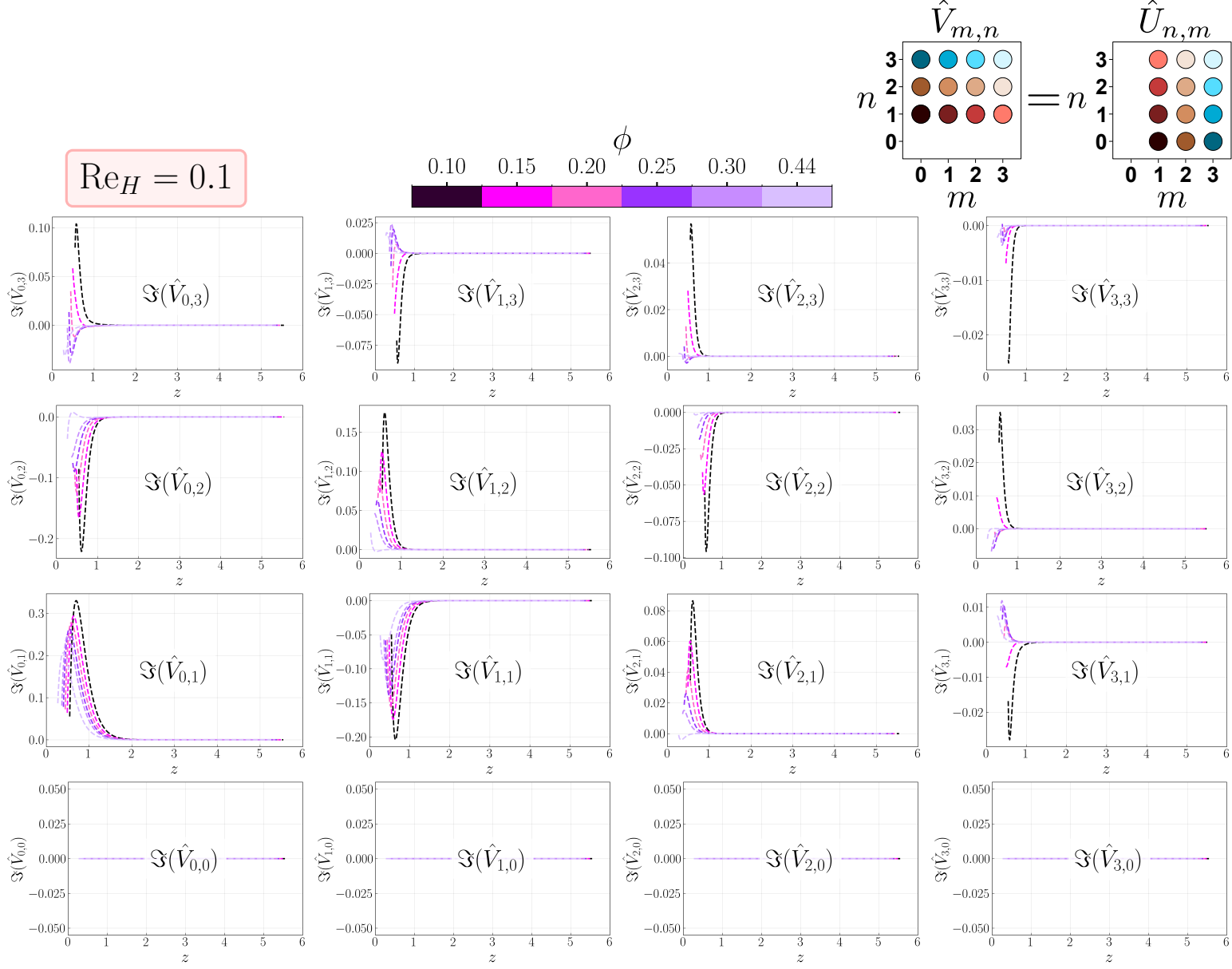


Figure B.24: Streamwise evolution of the imaginary part $\mathfrak{I}(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.1$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

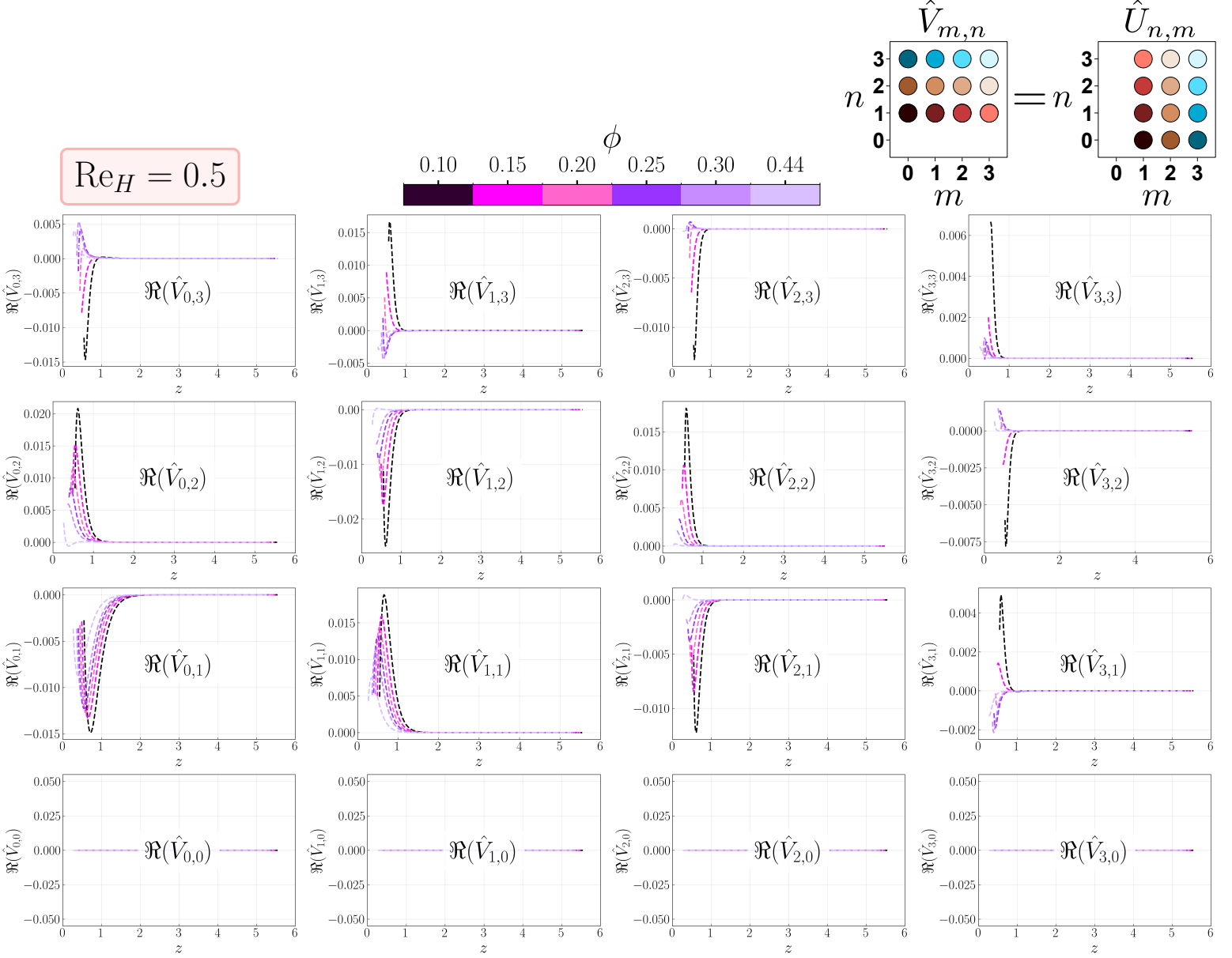


Figure B.25: Streamwise evolution of the real part $\Re(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.5$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

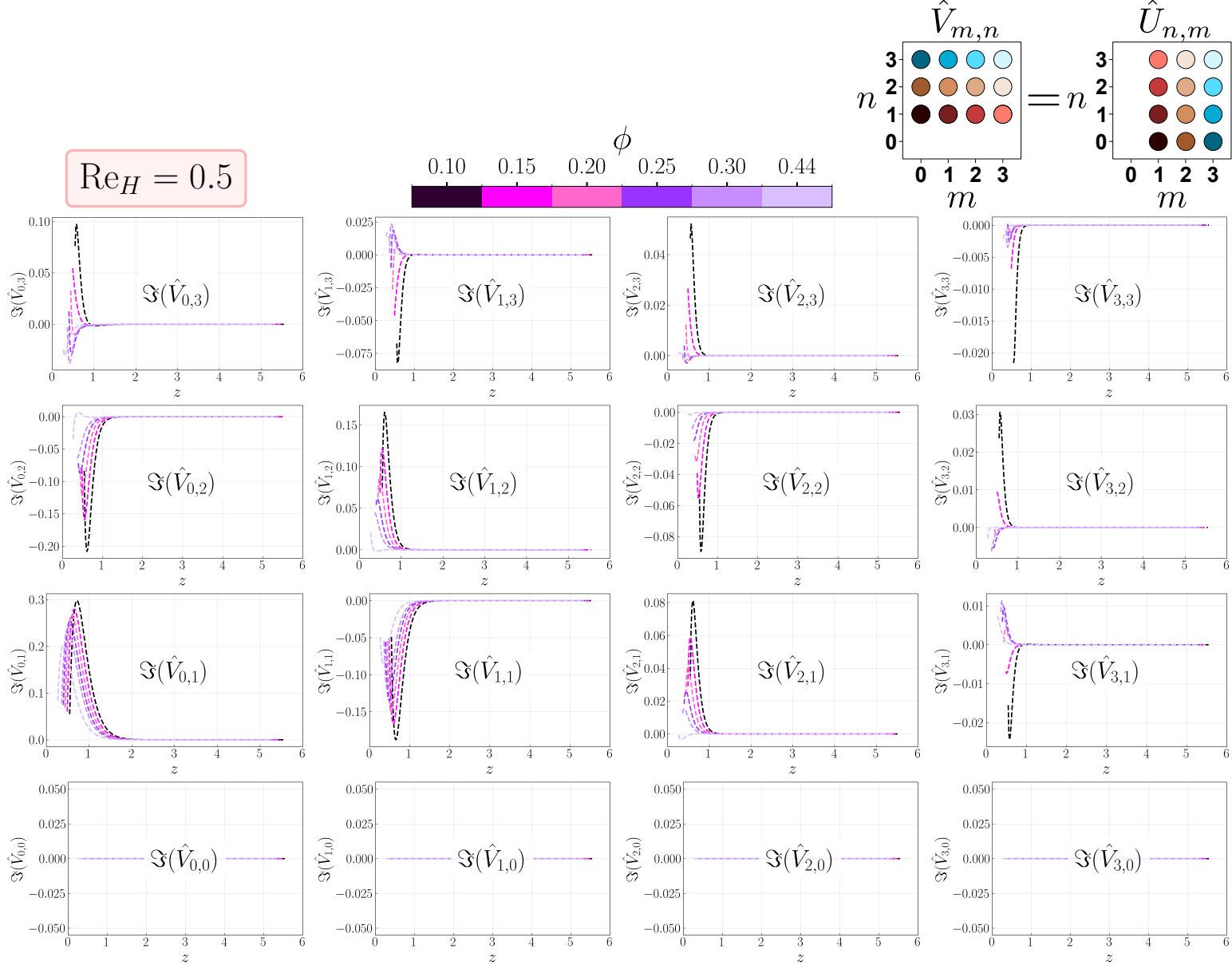


Figure B.26: Streamwise evolution of the imaginary part $\Im(\hat{V}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.5$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.